

Review problems

Problem 1. Assume, the observations $X_1, X_2, \dots, X_n$ are iid. normal distributed random variables with unknown mean θ . You observe $n = 16$ many variables with the empirical mean 1.45 and a sample variance of 0.51^2 .

- a) Determine a 90% two-sided confidence interval for the mean.
- b) How can we decide on the hypothesis $H_0 : \mu = 2$ vs $H_1 : \mu \neq 2$ on the significance level 10%, using just the answer for part a) and no additional computations?
- c) Now assume that, instead of using the sample variance, you know that the variance of the observed random variables $X_1, \dots, X_n$ is 0.5^2 ? What is the confidence of the confidence interval $[5/4, 13/8]$?

Problem 2. A random variable X is characterized by a normal density with mean $\mu_0 = 20$ and variance $\sigma_0^2 = 16$ (hypothesis H_0), or mean $\mu_1 = 25$ and variance $\sigma_1^2 = 25$ (hypothesis H_1). We want to test H_0 against H_1 , using four sample values x_1, x_2, x_3, x_4 and a rejection region of the form

$$R = \{x | x_1 + x_2 + x_3 + x_4 > \gamma\}$$

for some scalar γ . Determine the value of γ so that the probability of false rejection is 0.05. What is the corresponding probability of the false acceptance?

Problem 3. based on 8.6-7 Hogg. Let $X_1, \dots, X_{10}$ be a random sample from Poisson distribution with mean μ .

- a) Show that a uniformly most powerful critical region for testing $H_0 : \mu = 0.5$ against $H_1 : \mu > 0.5$ can be defined using sample mean statistic.
- b) Find a general formula for all uniformly best critical regions.
- c) Find a uniformly most powerful critical region of the size $\alpha = 0.068$. Hint: if $X_i \sim Poi(\mu)$ and independent, then $\sum_{i=1}^m X_i \sim Poi(m\mu)$
- d) Sketch the power function of this test.

Problem 4. Can p -value of some test be equal to 0.5? 2? -1?

Suppose p -value is equal to 0.3 and the probability to reject H_0 under some decision rule is 0.5. Do we accept or reject H_0 if it is additionally known that the probability that H_0 is false is 0.1? Explain why only the first sentence is not enough to give a conclusive answer.

Problem 5. Consider a regression model $Y_i = \beta x_i + \varepsilon_i$ for $i = 1, \dots, n$, where ε_i are independent and $N(0, \sigma^2)$. It is a version of a simple regression model with zero α , namely, when given $X = 0$, mean of Y is zero. Points x_i are assumed to be constants (not random variables).

Derive the maximum likelihood estimators, $\hat{\beta}$ and $\hat{\sigma}^2$, of β and σ^2 under this model. Do not use the known formulas for a simple regression problem. Show all the steps. What is the distribution of Y_i ? Of $\hat{\beta}$?

Problem 6. Is my coin fair? You flip a coin 200 times. It comes up head 85 many times and tails 115 many times. Using the Chi-Square test, do you accept the hypothesis that your coin is fair at the 5% significance level? Report an interval for the p-value.

Problem 7. You have a coin that shows head (i.e. $X = 1$) with probability $\theta \in [0, 1]$ and tail (i.e. $X = 0$) with probability $1 - \theta$. You do not know the value of θ but you assume that Θ is distributed according to the prior distribution

$$f_{\Theta}(\theta) = \begin{cases} c\theta, & \text{if } 0 \leq \theta \leq 1. \\ 0, & \text{else.} \end{cases}$$

Assume you observe tail in the next toss, what is the posterior distribution of θ ? How would you estimate success probability then? (Hint: what is the MAP estimate of θ ?)

Problem 8. Let $X_1, \dots, X_n$ be a random sample from the distribution with the density function $f(x; \theta) = 0.5e^{-|x-\theta|}$, $-\infty < x < \infty$ and $-\infty < \theta < \infty$. Find maximum likelihood estimator θ .

See also the hint in the exercise 6.4-8 of the textbook.

Problem 9. A sample size of $n = 100$ is taken from the production of light switches. A quality control engineer discovered that 10 switches does not meet the quality standard.

a) Suppose that another 500 switches are to be produces using the same technology. What are minimal and maximal amounts of defective switches should we expect with 95% confidence?

Hint: Use confidence interval for the defect rate of the production (=proportion of bad switches) to answer this question.

b) Estimate the confidence (=probability) that among the next 500 switches will be at most 60 defective switches.

Problem 10. On some magical island, there were elves, dwarfs, orcs and goblins. A researcher arrives to the island and decides to interview all four spices about their history and traditions. She invites 10 representatives of each folk for an interview. However, she soon finds out that at least some of the island inhabitants are quite mean and aggressive and don't interview well (whereas the rest are quite nice and friendly). So, the researcher wonders: are there more and less aggressive spices on the island? She records the data she got so far with in the following table:

	nice	not nice
elves:	3	7
dwarfs:	5	5
orcs:	4	6
goblins:	8	2

How can she test on the significance level 10% a hypothesis that being nice is a spices independent feature?

Specify what test you will use. Show all the steps. Make a conclusion (whether the researcher should accept or reject her hypothesis about independence).